MILITARY SPECIFICATION

RECEIVER-TRANSMITTER RT-1183( )/GRA-114

This specification is approved for use by Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Receiver-Transmitter RT-1183( )/GRA-114 which serves as the Command Post component of Radio Data Link, Sound Ranging AN/GRA-114 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallets, Materials-handling (General Construction Requirements)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock Container Grade and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and Connectors)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	Preservation-packaging, Methods of
MIL-S-3950	Switches, Toggle, General Specification for
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging

STANDARDS

FEDERAL

FED-STD-356	Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspections by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch X 48 Inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-A-678031	Burn-in Cycle
SM-A-865229	Performance Test Procedures

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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3 (see 6.2(g)).

3.2 Construction. The Receiver-Transmitter RT-1183 ( )/GRA-114 (Command Post) shall be constructed in accordance with the requirements of this specification and model.

3.2.1 Knobs. Knobs on code switches shall be provided with key-ways to insure proper orientation on the switch shafts.

3.2.2 Transmit filters. High-pass and low-pass filters required for duplex transmitter operation shall be installed on their respective command transmitters.

3.2.3 Battery cable. The battery cable shall be demountable and use a 90° connector. Cable shall be clamped during operation. Mounting fixtures shall be provided in the lid to permit damage-free and convenient storage during transit.

3.2.4 Data START/STOP. Depressing either data command button shall generate a code train of sufficient duration to insure acquisition of the command by the controlled equipment without requiring repetitive actions by the operator.

3.2.5 Printed circuit boards. Where feasible printed circuit boards shall be provided with pull-out flanges. Flanges shall be labeled.

3.2.5.1 Test points. Test points shall be provided on the printed circuit boards listed below. Test points shall be properly labeled and located so as to facilitate testing without removal of the boards from the chassis wherever feasible.

a. CP Receiver Logic Board
b. CP Encode Board
c. CP Decode Board

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3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply.

3.3.1 Toggle switches. Toggle switches, where employed, shall be full size in accordance with MIL-S-3950.

3.4 Finish. All surfaces of the RT-1183 requiring a protective coating shall be finished in accordance with MIL-F-14072 (see 4.4).

3.5 Marking.

3.5.1 General. Marking shall be in accordance with MIL-M-13231 (see 4.4). Front panel markings shall be group I as described in that specification. Multiple P.C board enclosures shall be marked to show the functions and positions of their interior contents.

3.5.2 Visibility. Whenever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. Each RT-1183 shall be serial-numbered (see 4.4).

3.6 Interchangeability. Like units, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.6).

3.7 Burn-in. The RT-1183 shall meet the requirements of this specification after being subjected to the burn-in cycle of Drawing SC-A-678031 (see 4.7).

3.8 Performance requirements.

3.8.1 Transmitter.

3.8.1.1 Power output. The minimum power output as measured at the external fitting for the antenna coaxial cable shall be as follows:

Temp	Battery (Volts)	Power Out (W)
-40°C to 65.6°C (-40°F to 150°F) 1/	12-15 1/	3.15
10°C to 65.6°C (50°F to 150°F)	10	2.25
10°C to -40°C (50°F to -40°F)	10	2.25

(see 4.9)

1/ Equipment to be capable of continuous operation under these conditions.

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3.8.1.2 Efficiency. The efficiency shall be at least 27 percent (see 4.9).

3.8.1.3 Meter reading. The panel meter reading shall show at least the minimum signal marking on the meter scale when transmitter is operating from a 10.0 VDC source (see 4.9).

3.8.1.4 Frequency. The frequency shall be within .007 percent of nominal channel frequency (see 4.9).

3.8.1.5 Modulation. The deviation shall not be greater than +10 KHz nor less than +7 KHz when a 1.0 mv, 1000 Hz, signal is applied at the handset connector (see 4.9).

3.8.1.6 Audio frequency response. The frequency response shall meet the requirements of Table I (see 4.9).

TABLE I. Audio frequency response.

Frequency (HZ)	Response (dB)
150	-3 +1.5
300	-1 ±1.0
600	0 ±1.0
1000	0 reference
3500	-3 +2.0
5000	-5 min.

3.8.1.7 Deviation limiting. The deviation shall not be greater than +19.0 KHz when a 1.0 KHz signal of 10 mv is applied to the input of the handset connector (see 4.9).

3.8.1.8 Audio distortion. The audio distortion shall be less than 5 percent with a carrier deviation of ±8 KHz at a frequency of 1000 Hz (see 4.9).

3.8.1.9 Encode deviation. The encode deviation shall be 8 KHz ±2 KHz for a START or STOP command transmission (see 4.9).

3.8.1.10 Command indicator. Depressing the DATA START button shall activate an indicator light which shall remain lit until a STOP command is issued (see 4.9).

3.8.1.11 Duplexing loss. Duplexing filters shall not result in a power loss in excess of 2dB as measured between the R.F. output terminals of the transmitter and the external antenna connector of the RT-1183 (see 4.9).

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3.8.2 Receiver, code channel.

3.8.2.1 Code receiver sensitivity. The Radio-Frequency carrier level required for code acceptance of any code selected by the operator shall not be greater than 1.0 uv. Not more than a single train of code groups shall be necessary for triggering action (see 4.9).

3.8.2.2 Duplexing desensitization. The receiver desensitization due to concurrent operation of the local transmitter in RT-1183 shall not exceed 2 dB as compared to the corresponding response to input signals exhibited by the receiver when the RT-1183 transmitter is in standby condition (see 4.9).

3.8.2.3 Command channel capture ratio. The capture ratio for command signals shall not exceed 3 dB, when measured at 5 mv, and 100 mv signal strength (see 4.9).

3.8.3 Receivers, voice channel.

3.8.3.1 Sensitivity. The sensitivity shall be such that a 0.5 uv modulated signal applied to the receiving antenna input terminal of the RT-1183 shall produce a signal-plus-noise-to-noise ratio of at least 10 dB in any voice channel receiver (see 4.9).

3.8.3.2 Front end overload. An interfering signal, 120 dB above 1 micro volt and removed $\pm$15 MHz from the desired frequency, shall not cause failure of any part of RT-1183.

3.8.3.3 Limiting. Not more than 1.0 uv of input signal shall be required to achieve full limiting. The limiting characteristics shall be such that the audio power output shall not vary more than $\pm$3 dB at RF input levels of 1.0 to 100,000 uv (see 4.9).

3.8.3.4 S-Meter calibration. With a 1.0 uv signal applied to the receiving antenna input terminal of RT-1183, the panel meter reading shall not be less than minimum signal scale marking. The meter reading shall increase but be less than full scale when the RF level is increased to 5 uv (see 4.9).

3.8.3.5 Distortion. Audio distortion at 5 mw output (phones) or 400 mw, (speaker) shall not exceed 5 percent (see 4.9).

3.8.3.6 Audio output power. The output level at 12.0 VDC power supply shall not be less than 2.24 vrms across 1000 ohms impedance and the output level at 10 VDC shall not be less than 1.73 vrms (see 4.9).

3.8.3.7 Loudspeaker power. When tested in accordance with 4.9, the voltage developed across the voice coil of the speaker shall be at least 1.79 vrms.

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3.8.3.8 Voice frequency response. The frequency response shall be in accordance with Table II (see 4.9).

TABLE II. Voice frequency response.

Frequency (Hz)	Output (dB)
150	-3 +1.5
300	-1 ±1.0
600	0 ±1.0
1000	0 reference
3500	-3 +2.0
5000	-6 min.

3.8.4 Receiver, data.

3.8.4.1 Data gain. The data signal level shall be 7.0 vrms +2.0, - 1.0 vrms when tested at a carrier signal level of 100 uv deviated ±10 KHz with a 25 Hz modulating signal (see 4.9).

3.8.4.2 Data distortion. The data distortion shall not exceed 3 percent for data frequencies of 5 to 100 Hz (see 4.9).

3.8.4.3 Data frequency response. The frequency response of the data channels shall comply with Table III (see 4.9).

TABLE III. Data frequency response

Frequency (Hz)	Output (dB)
5	-1, +2.0, -1.0
10	-.5 ± 1.0
25	0 reference
75	0 +1.0
100	0 ±1.0
150	-1.5 max
300	-5.0 max
600	-3.0 min
1000	-5.0 min
2000	-16.0 min

3.8.4.4 Dynamic range. The output level shall not exceed 7 mvrms, when a 100 uv CW signal is applied to the receiver antenna terminals. (See 4.9).

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3.8.4.5 Receiver, data, sensitivity. The receiver data sensitivity shall be such that a 0.5 uv modulated signal applied to the receiving antenna terminal of the RT-1183 shall produce a signal-plus-noise-to-noise ratio of at least 10 dB from any data receiver measured at the recorder terminals (see 4.9).

3.8.4.6 Receiver, data, transient immunity. Connecting the receiver output terminals to the input terminals of the data recorder during periods when there is high voltage present, shall not result in damage to the receivers. The output circuits in the data receiver shall be capable of withstanding transients introduced as a result of deliberate or accidental shorting of the recorder terminals (see 4.9).

3.8.4.7 Start-up delay. All components required in the reception of data signals shall be operational and fully stabilized within 300 ms of the reception of a Data Start signal (see 4.9).

3.8.5 Telemetry operation. The telemetry signals shall provide an indication of the condition of the microphone M-161 batteries and the RT-1184 internal batteries. The presence of a carrier signal on any receiver channel shall cause a steady light to glow on the active channel indicator for good batteries and cause a blinking light for weak batteries (see 4.9).

3.8.6 Recorder control operation. The outpost terminals shall permit the flow of current whenever a properly coded RF signal of 1.0 uv or more is applied to the command receiver antenna terminals (see 4.9).

3.8.7 Communications indicator. A visual alarm shall alert the operator to the presence of carrier signals on any receiver channel(s). Activation of a channel shall cause an incandescent lamp to be illuminated and shall energize the associated audio circuit. Only the receiving channel which is tuned to the incoming frequency shall provide a visual indication of activity. The lamp shall remain lit until the transmission from the remote unit has been completed, after which all affected circuits shall automatically revert to standby condition. The lamps shall be easily demountable for replacement purposes (see 4.9).

3.8.8 Phase shift. The time differential between data receiver channels due to phase shifts shall not exceed 0.5 ms (see 4.9).

3.9 Service conditions. The equipment shall meet the service conditions below.

3.9.1 Temperature.

3.9.1.1 High temperature. During testing as specified in 4.10.1.1, the RT-1183 shall meet the following requirements:

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Paragraph	Measurement
3.8.1.1	Power Output
3.8.1.4	Frequency
3.8.2.1	Code receiver sensitivity
3.8.3.1	Sensitivity
3.8.3.5	Distortion
3.8.3.6	Audio output power
3.8.4.7	Start-up delay

3.9.1.2 Low temperature. During testing as specified in 4.10.1.2, the RT-1183 shall meet the following requirements:

Paragraph	Measurement
3.8.1.1	Power output
3.8.1.4	Frequency
3.8.2.1	Code receiver sensitivity
3.8.3.1	Sensitivity
3.8.3.5	Distortion
3.8.3.6	Audio output power
3.8.4.7	Start-up delay

3.9.2 Humidity. The RT-1183 shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet the following requirements during the operating times specified in 4.10.2.

Paragraph	Measurement
3.8.1.1	Power output
3.8.1.4	Frequency
3.8.2.1	Code receiver sensitivity
3.8.3.1	Sensitivity
3.8.3.5	Distortion

3.9.3 Immersion. The RT-1183 shall show no evidence of leakage after it is tested as specified in 4.10.3.

3.9.4 Dust. After testing as specified in 4.10.4, the RT-1183 shall meet full specification performance for the following measurements.

Paragraph	Measurement
3.8.1.1	Power output
3.8.3.1	Sensitivity

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3.9.5 Vibration, bounce, and shock.

3.9.5.1 Bounce and shock. The RT-1183 shall meet the requirements of Table IV.

TABLE IV. - Bounce and shock.

Requirement	Test Paragraph	Performance After Test
Bounce	4.10.5.1	Specified performance 1/ No physical damage except minor surface abrasion
Shock, Bench-Handling	4.10.5.2	Specified performance 1/ No physical damage
Shock; Drop	4.10.5.3	Specified performance 1/ Any physical damage shall be minor

1/ The RT-1183 shall meet specified performance for the following:

Paragraph	Measurement
3.8.1.1	Power output
3.8.1.2	Efficiency
3.8.1.4	Frequency
3.8.2.1	Code receiver sensitivity
3.8.3.1	Sensitivity
3.8.3.5	Distortion
3.8.3.6	Audio output power

3.9.6 Rain. After testing as specified in 4.10.6, there shall be no evidence of water penetration into the interior of the transit case.

3.9.7 Salt fog. After the salt fog test of 4.10.7, the RT-1183 when examined visually, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish of metal surface. Electrical operation is not required.

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3.9.8 Fungus. After fungus test of 4.10.8, the RT-1183 shall be visually examined and shall reveal no evidence of viable fungus on the surface.

3.10 Electromagnetic interference (EMI) characteristics. The RT-1183 shall meet the emission and susceptibility requirements for MIL-STD-461 for class I equipments. Test setups and test procedures shall conform to the measurement techniques of MIL-STD-462 (see 4.11). The following requirements are applicable:

Transmit Mode	Receive	Mode
CE06 3/	CS03	CE06
RE02 1/	CS04 2/	RE02 1/
RS03 1/	CS05	RS03 1/

1/ The upper frequency of the test shall be 1000 MHz.

2/ All responses shall be 96 dB down except for a total of five (5) which shall not be less than 80 dB down, when tested in accordance with MIL-STD-462.

3/ a. The equipment shall meet the spurious and harmonic output requirements of MIL-STD-461A, Notice 4, except for a total of three responses which may be between 70 and 80 dB down. No response will be less than 70 dB down relative to the fundamental power in the medium power mode.

b. The equipment shall meet the spurious and harmonic output requirements of MIL-STD-461A, Notice 4, except for a total of three responses which may be between 70 and 80 dB down. No response will be less than 70 dB down. The limit at the second harmonic shall be relaxed by 5 dB when operated in the high power mode.

3.11 Weight. The RT-1183 shall not exceed 38 lbs, less crystal modules, antenna amp., and handset. (See 4.12).

3.12 Reliability. The RT-1183( )/GRA-114 shall have a specified MTBF (θ°) of 440 hours. The test specified in paragraph 4.14 shall be conducted to determine compliance with this reliability requirement.

3.13 Workmanship. The RT-1183 shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs in MIL-P-11268 (see 4.4):

General requirements for plastic material and plastic parts
Wiring and cabling, including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable

Soldering and brazing
Cleaning of units
General requirements for securing of parts
Cleaning prior to welding
Controls
Self tapping screws
Cleaning after welding
Printed wiring assemblies

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspections requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). (Does not include packaging.)

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.8).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish four (4) first article units of the Receiver Transmitter RT-1183( )/GRA-114 for Group A, B. C and D testing.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in Table V, and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test report shall be as required in the contract or purchase order.

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TABLE V. - First article inspection.

Inspections	Req Para	Test Para	Unit 1	Unit 2	Unit 3	Unit 4
1. Inspections covered by subsidiary documents. 1/	3.3 3.4 3.5 3.13	4.4	Inspection to be performed all units.			
2. Burn-in 1/	3.7	4.7	Tests to be performed all units.			
3. Group A inspection 1/	See Table VI		Tests to be performed on all units.			
4. Group B inspection 1/	See Table VII		Tests to be performed on all units.			
5. Group C inspection 1/						
Front end overload	3.8.3.2	4.9		2		
High temperature	3.9.1.1	4.10.1.1	1			
Low temperature	3.9.1.2	4.10.1.2	2			
Humidity	3.9.2	4.10.2	4			
Immersion	3.9.3	4.10.3	6			
Dust	3.9.4	4.10.4		4		
Bounce	3.9.5.1	4.10.5.1		5		
Shock, bench-handling	3.9.5.1	4.10.5.2		6		
Shock-drop	3.9.5.1	4.10.5.3	5			
Rain	3.9.6	4.10.6	3			
Salt fog	3.9.7	4.10.7		7		
Fungus	3.9.8	4.10.8			1	
EMI	3.10	4.11		3		
Weight	3.11	4.12		1		
6. Group D inspection	3.12	4.14				1

1/ Inspections 1 to 4 shall be performed in the order shown before subjecting the equipments under test to any other inspection requirements.

2/ The numbers in the unit columns in the table, for Inspection 5 specify the order of testing for the indicated units.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Parts, materials, and processes; general	3.3
Finish	3.4
Marking	3.5
Workmanship	3.13

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.4, and 4.7. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspection in Table VI. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contractor or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels and AQL's indicated in Table VI.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

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TABLE VI. Group A inspection.

Inspection	Req Para	Test Para	AQL Major	AQL Minor
Visual & Mechanical	3.13	4.13	1%	4%
Electrical				1/
Power output	3.8.1.1	4.9	1%	
Meter reading	3.8.1.3	4.9	1%	
Modulation	3.8.1.5	4.9	1%	
Audio frequency response	3.8.1.6	4.9	1%	
Audio distortion	3.8.1.8	4.9	1%	
Encode deviation	3.8.1.9	4.9	1%	
Command indicator	3.8.1.10	4.9	1%	
Code receiver sensitivity	3.8.2.1	4.9	1%	
Sensitivity	3.8.3.1	4.9	1%	
Limiting	3.8.3.3	4.9	1%	
S-Meter calibration	3.8.3.4	4.9	1%	
Distortion	3.8.3.5	4.9	1%	
Audio output power	3.8.3.6	4.9	1%	
Loudspeaker power	3.8.3.7	4.9	1%	
Telemetry operation	3.8.5	4.9	1%	
Recorder control operation	3.8.6	4.9	1%	
Data distortion	3.8.4.2	4.9	1%	
Receiver, data, sensitivity	3.8.4.5	4.9	1%	
Communications indicator	3.8.7	4.9	1%	

1/ All electrical defects are major

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table VII and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sample plans, for the AQL's listed in Table VII, shall be as follows:

AQL	Inspection level
4%	S-4
6.5%	S-4

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4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

TABLE VII. Group B inspection.

Inspection	Requirement Paragraph	Test Paragraph	AQL
Interchangeability	3.6	4.6	6.5%
Electrical			
Efficiency	3.8.1.2	4.9	4%
Frequency	3.8.1.4	4.9	4%
Deviation limiting	3.8.1.7	4.9	4%
Duplexing desensitization	3.8.2.2	4.9	4%
Command channel capture ratio	3.8.2.3	4.9	4%
Voice frequency response	3.8.3.8	4.9	4%
Data gain	3.8.4.1	4.9	4%
Data frequency response	3.8.4.3	4.9	4%
Dynamic range	3.8.4.4	4.9	4%
Receiver, data transient immunity	3.8.4.6	4.9	4%
Start-up delay	3.8.4.7	4.9	4%
Phase shift	3.8.8	4.9	4%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table VIII. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection. Three (3) samples of RT-1183 shall be selected at random, one (1) sample for each subgroup in Table VIII from the first thirty (30) units or fraction thereof produced. The EMI and fungus tests shall be performed only on the first production lot. Thereafter, two (2) samples of RT-1183 shall be selected at random from every thirty (30) units or fraction thereof greater or equal to 15 units, one unit for each subgroups 1 and 2 of Table VIII. If the fraction is less than 15, the selection of Group C samples for the preceding test shall include units from the remaining fraction.

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4.5.3.2 Group C failures. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.3.3 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

Table VIII. Group C inspection.

Inspection	Requirement Paragraph	Test Paragraph
Subgroup 1		
High temperature	3.9.1.1	4.10.1.1
Low temperature	3.9.1.2	4.10.1.2
Immersion	3.9.3	4.10.3
Rain	3.9.6	4.10.6
Shock-drop	3.9.5.1	4.10.5.3
Humidity	3.9.2	4.10.2
Subgroup 2		
Dust	3.9.4	4.10.4
Bounce	3.9.5.1	4.10.5.1
Shock, bench-handling	3.9.5.1	4.10.5.2
Salt fog 1/	3.9.7	4.10.7
Duplexing loss	3.8.1.11	4.9
Front end overload	3.8.3.2	4.9
Subgroup 3		
EMI	3.10	4.11
Fungus 1/	3.9.8	4.10.8

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after their inspection before proceeding with subsequent inspection.

4.5.4 Group D inspection. This inspection shall consist of the test specified in 4.14 and shall be performed on units which have been subjected to and have passed Group A and Group B inspections.

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4.5.4.1 Sampling for Group D inspection. One (1) unit shall be selected at random from each 6 months production and subjected to the test of paragraph 4.14.

4.5.4.2 Non-compliance. In case of Group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed.

4.5.4.3 Reinspection of Group D sample units. Unless otherwise specified, sample units which have been subjected to, and have passed Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.6 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.7 Burn-in. Each RT-1183 submitted for acceptance shall be subjected to burn-in in accordance with Drawing SC-A-678031 (see 3.7).

4.8 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

4.9 Performance characteristics. The performance characteristics of Receiver-Transmitter RT-1183( )/GRA-114 shall be tested in accordance with the Performance Test Procedure, Drawing SM-A-865229. Paragraph titles in the test procedure correspond to the requirements paragraphs of the specification.

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Title	Requirement paragraph
Power output	3.8.1.1
Efficiency	3.8.1.2
Meter reading	3.8.1.3
Frequency	3.8.1.4
Modulation	3.8.1.5
Audio frequency response	3.8.1.6
Deviation limiting	3.8.1.7
Audio distortion	3.8.1.8
Encode deviation	3.8.1.9
Command indicator	3.8.1.10
Duplexing loss	3.8.1.11
Code receiver sensitivity	3.8.2.1
Duplexing desensitization	3.8.2.2
Command channel capture ratio	3.8.2.3
Sensitivity	3.8.3.1
Front end overload	3.8.3.2
Limiting	3.8.3.3
S-Meter calibration	3.8.3.4
Distortion	3.8.3.5
Audio output power	3.8.3.6
Loudspeaker power	3.8.3.7
Voice frequency response	3.8.3.8
Data gain	3.8.4.1
Data distortion	3.8.4.2
Data frequency response	3.8.4.3
Dynamic range	3.8.4.4
Receiver, data, sensitivity	3.8.4.5
Receiver, data, transient immunity	3.8.4.6
Start-up delay	3.8.4.7
Telemetry operation	3.8.5
Recorder control operation	3.8.6
Communications indicator	3.8.7
Phase shift	3.8.8

4.10 Service condition. The RT-1183 shall be tested in accordance with the following conditions.

4.10.1 Temperature.

4.10.1.1 High temperature. The RT-1183 shall be subjected to the test of Method 501.1, Procedure II of MIL-STD-810. For Step 4, the chamber temperature shall be 71°C (160°F). For Step 7, the chamber temperature shall be adjusted to 65.6°C (150°F). (This includes effects of solar radiation) (see 3.9.1.1).

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4.10.1.2 Low temperature. The RT-1183 shall be subjected to the test of Method 502.1, Procedure I of MIL-STD-810. The storage temperature (Step 2) shall be -45.6°C (-50°F), and shall be maintained until temperature stabilization of the test item is reached. Low operating temperature (Step 4) shall be -12°C (10°F) (see 3.9.1.2).

4.10.2 Humidity. The RT-1183 shall be subjected to the test of Method 507.1, Procedure III of MIL-STD-810. In Step 7 of Procedure III, the panel shall be opened and protective covers removed from the shield compartments, so as to expose to the chamber environment the normally sealed areas. In Steps 6 and 8, measurements shall be taken during the last test cycle (see 3.9.2).

4.10.3 Immersion. The RT-1183 with the cover secured shall be subjected to the test of Method 512.1, Procedure I of MIL-STD-810 (see 3.9.3).

4.10.4 Dust. The RT-1183 with the cover secured shall be subjected to the test of Method 510.1, Procedure I of MIL-STD-810 (see 3.9.4).

4.10.5 Vibration, bounce and shock.

4.10.5.1 Bounce. The RT-1183 shall be subjected to the test of Method 514.2 Procedure XI, Part 2 of MIL-STD-810. It shall undergo one hour of testing at each of the following temperatures: (1) Room ambient (2) -45°C(-50oF) (3) 71°C(160°F)

At each of these temperatures, RT 1183 shall be vibrated for ten minutes on each face. (See 3.9.5.1.)

4.10.5.2 Shock, bench-handling. The RT-1183 shall be subjected to the test of Method 516.2, Procedure V of MIL-STD-810 (see 3.9.5.1).

4.10.5.3 Shock, drop. The RT-1183 in transit configuration shall be subjected to the test of Method 516.2, Procedure II of MIL-STD-810 (see 3.9.5.1).

4.10.6 Rain. The RT-1183, with cover open shall be subjected to the test of Method 506.1, Procedure I of MIL-STD-810. During each 30 minute rain period, the wind shall be maintained for 15 minutes, commencing five minutes after the start of the period as specified in the procedure (see 3.9.6).

4.10.7 Salt fog. The RT-1183 with cover open shall be subjected to the test of Method 509.1, Procedure 1 of MIL-STD-810 (see 3.9.7).

4.10.8 Fungus. The RT-1183 shall be subjected to the test of Method 508.1, Procedure 1 of MIL-STD-810. Gasket sealed or nonhermetic sealed test items shall be opened during exposure and internal areas as well as external surfaces shall be sprayed with spore suspension. Operation of the equipment is not required (see 3.9.8).

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4.11 Electromagnetic interference characteristics. The RT-1183 shall be tested in accordance with MIL-STD-462 and meet requirements of 3.10.

4.12 Weight. The RT-1183 shall be weighed and shall meet the requirement of 3.11.

4.13 Visual and mechanical inspection. The RT-1183 shall be examined for the applicable defects listed in MIL-STD-252.

4.14 Reliability testing.

4.14.1 First article and quality conformance Group D inspection. To determine compliance with paragraph 3.12 during First Article and Quality Conformance inspection, the equipment shall be tested in accordance with Test Plan XXIV of MIL-STD-781. The specified MTBF for all reliability testing shall be 440 hours.

4.14.2 Failure definition. A failure of the RT-1183( )/GRA-114 occurs in the event of any one of the following:

a. Any 3 out of 8 Command Post Data/Voice Receivers fail to perform as specified during any 24 hour operating time period.

b. Both Command Post, Command/Voice Receivers fail to perform as specified during any 24 hour operating time period.

c. The Command Post transmitter or other Command Post component (excluding the receivers) fails to perform as specified.

4.14.3 Test conditions. The test shall be conducted under the conditions of Test Level C of MIL-STD-781 with the exception of no vibration. The test shall be conducted in accordance with the reliability test plan required in the contract.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or Commercial as specified (see 6.(2)(b)).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Receiver-Transmitter RT-1183 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Receiver-Transmitter RT-1183 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified.

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5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Receiver-Transmitter RT-1183( )/GRA-114. Each RT-1183 shall be individually unit packed method III as follows: place the required components within the space provided and secure the fasteners. Cushion the RT-1183 on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the container. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B.

5.1.2.1 Cleaning, drying, preservation application and unit packaging. Each unit shall be cleaned, dried, preserved and unit packed as specified in 5.1.1.

5.1.3 Commercial. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.(2)(b)). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Receiver-Transmitter RT-1183( )/GRA-114, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 X 24 inches or more and the weight exceeds 100 pounds, 3 X 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as

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cludes any event which causes departure from performance as ... defined a failure. This

specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.2 Level B. A quantity of Receiver-Transmitter RT-1183( )/GRA-114, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640, shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 X 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when: quantities per destination comprise a pallet load. A quantity of Receiver-Transmitter RT-1183( )/GRA-114, unit packed as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the packed load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film.

5.2.3 Commercial packing. Commercial packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

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6. NOTES

6.1 Intended use. The Receiver-Transmitter RT-1183( )/GRA-114 (Command Post) is used in conjunction with Receiver-Transmitter RT-1184( )/GRA-114 (Microphone Position) and Receiver-Transmitter RT-1185( )/GRA-114 (Sound Observer) to provide a radio data link for sound ranging applications. It provides voice and data communications with exterior units, and controls the initiation of data transmissions from the Microphone Positions.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, level B or commercial preservation and packing (see Section 5).

c. When rough handling test is required.

d. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Number of first articles to be submitted for approval (see 3.1).

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted and replaced by a letter identifying the particular design for example RT-1183W/GRA-114. The contractor shall should apply for nomenclature in accordance with applicable clause in the contract (see 1.1).

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DOCUMENT IDENTIFIER AND TITLE MIL-R-49111(EL) RECEIVER-TRANSMITTER RT-1183( )/GRA-114

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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